Sensing and the Imaging Process (III)

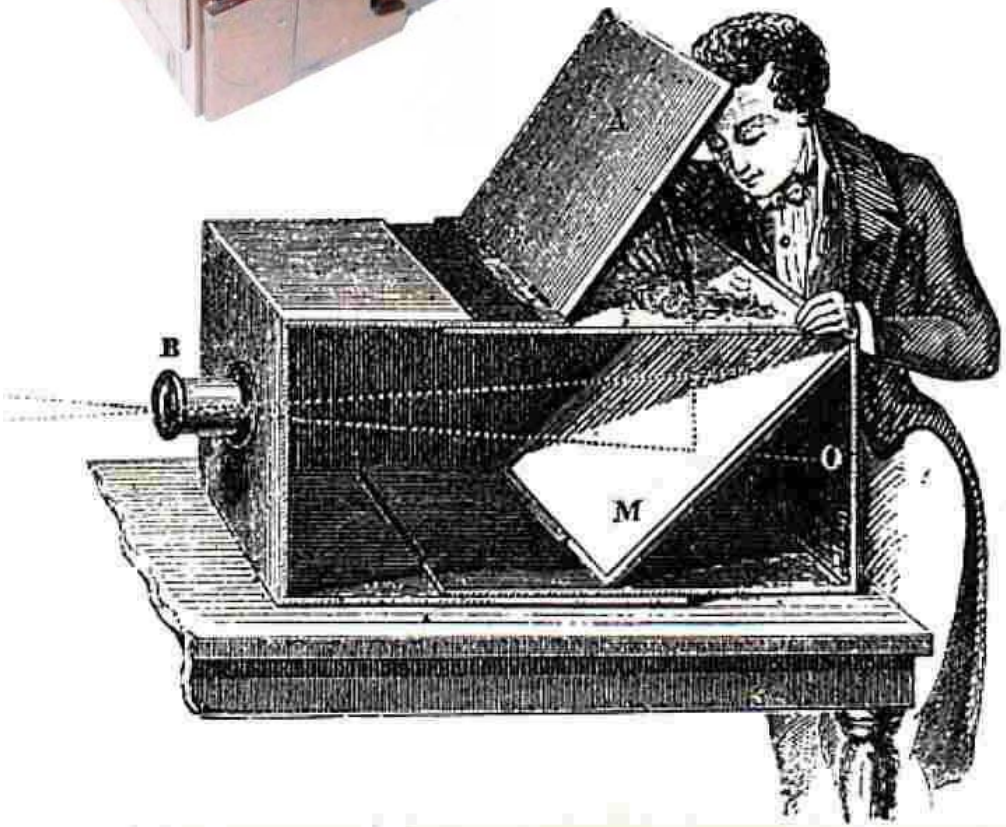
Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

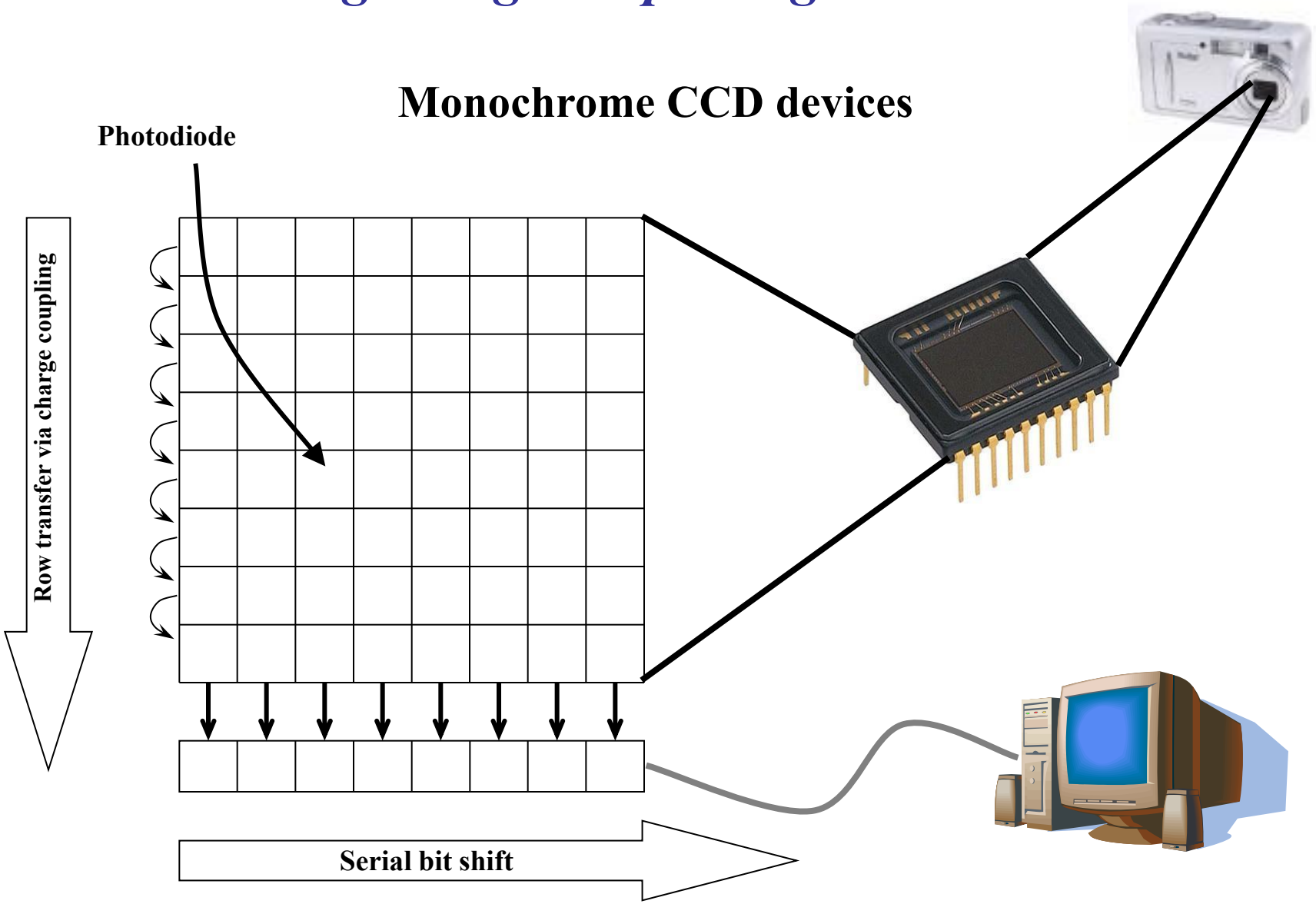
Ohad Ben-Shahar

Artificial light-capturing devices (a.k.a. cameras)



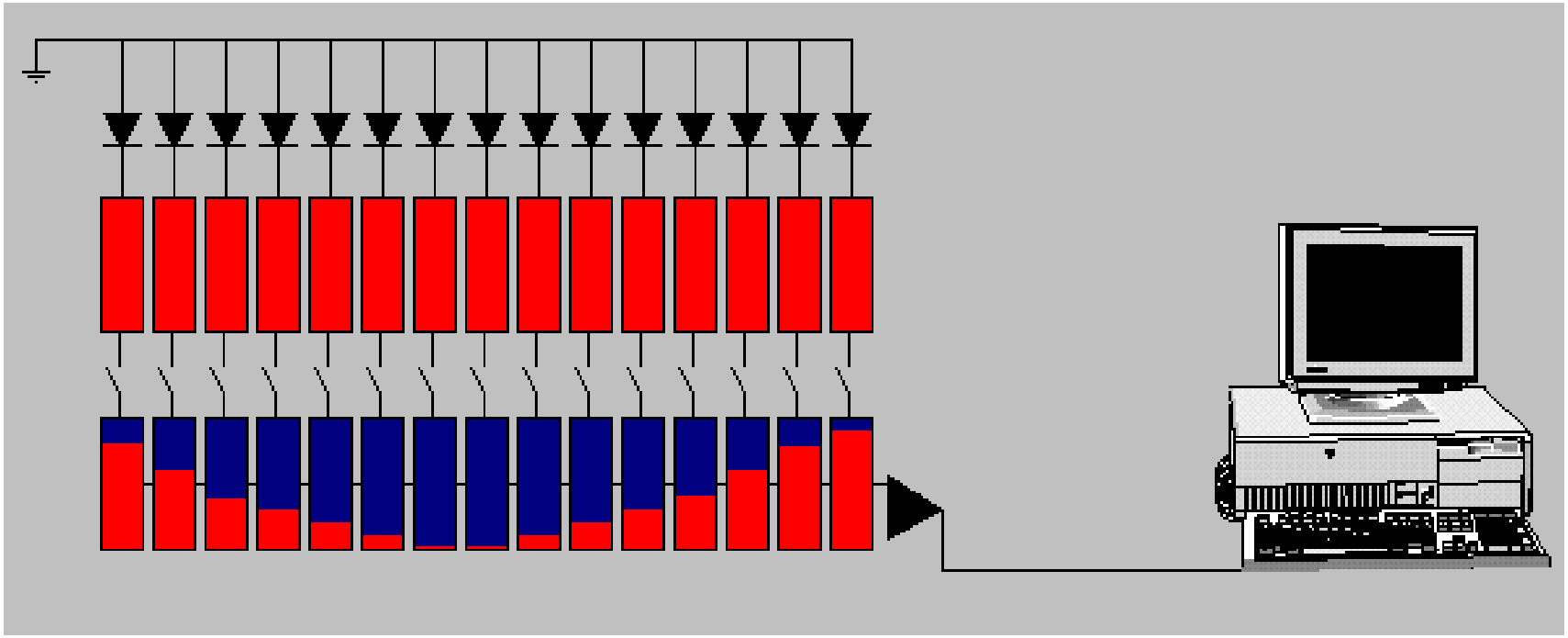
Digital light-capturing devices

Monochrome CCD devices



Digital light-capturing devices

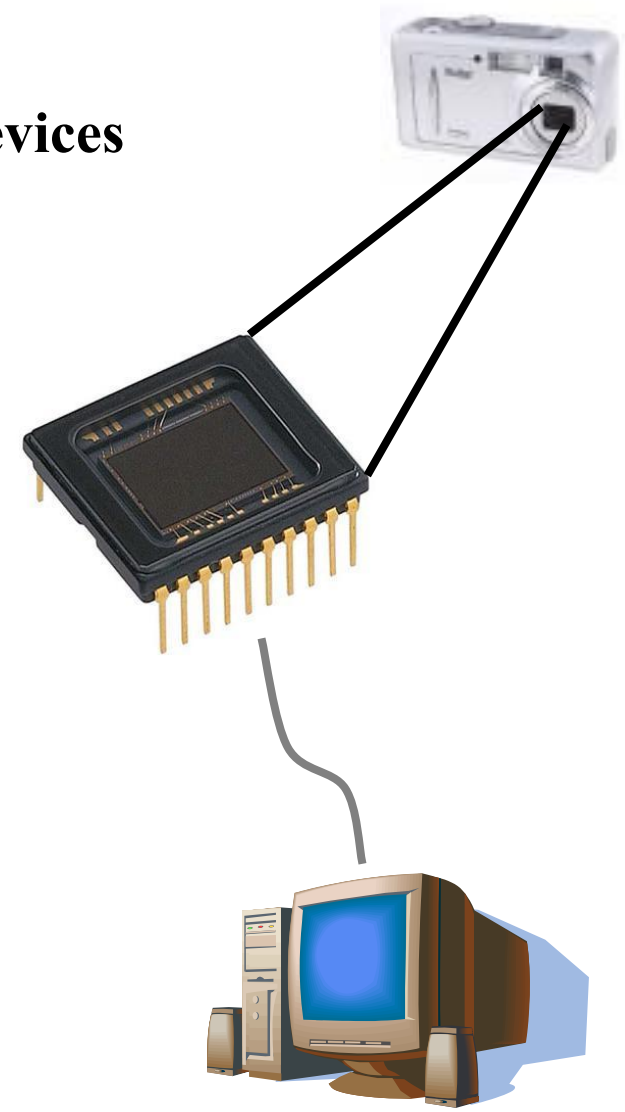
Monochrome CCD devices



Digital light-capturing devices

Monochrome CMOS devices

Property	CCD	CMOS
Image noise	Lower	Higher
electronics/pixel	Less	More
Design complexity	Lower	Higher
Exposure	Global shutter	Rolling shutter
quantum efficiency	Higher	Lower
Potential frame rate	Lower	Higher
Power consumed	Much more	Much less
Fabrication cost	Much higher	Much lower
Maturity	More	Less



Digital light-capturing devices

Monochrome CMOS devices

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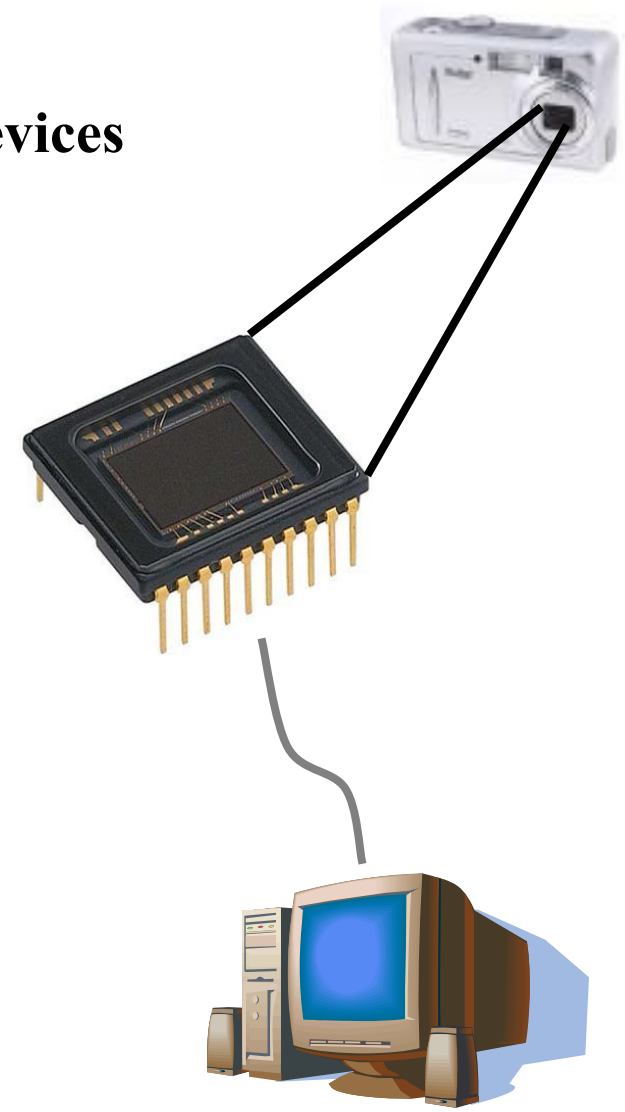

Digital light-capturing devices

Monochrome CMOS devi

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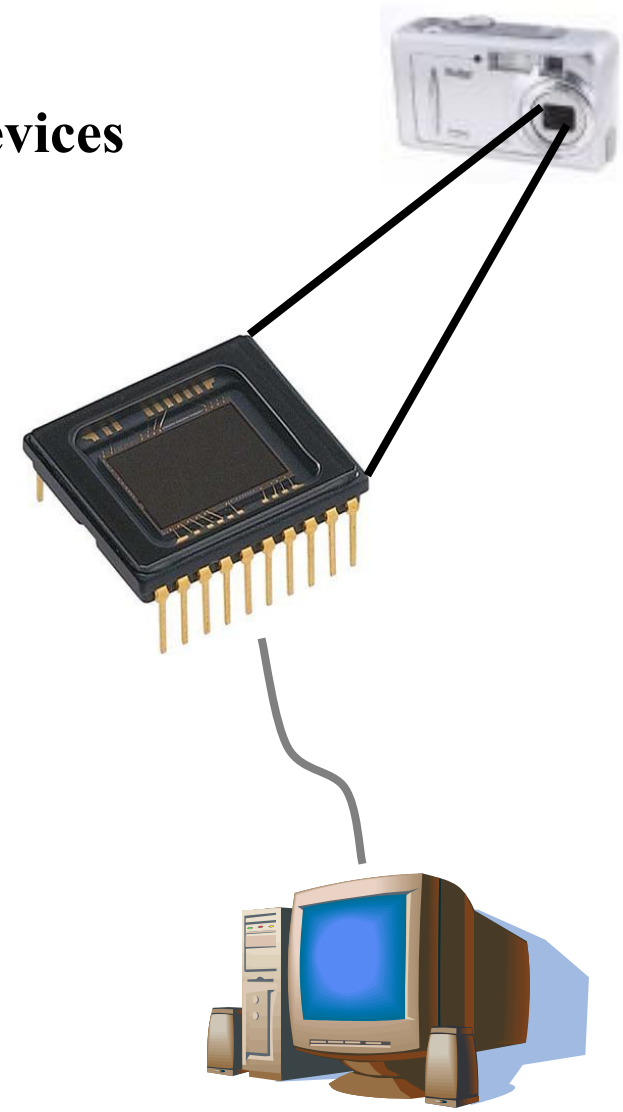
electronics/pixel	Less	More
Design complexity	Lower	Higher
Exposure	Global shutter	Rolling shutter
quantum efficiency	Higher	Lower
Potential frame rate	Lower	Higher
Power consumed	Much more	Much less

Digital light-capturing devices

Monochrome CMOS devices

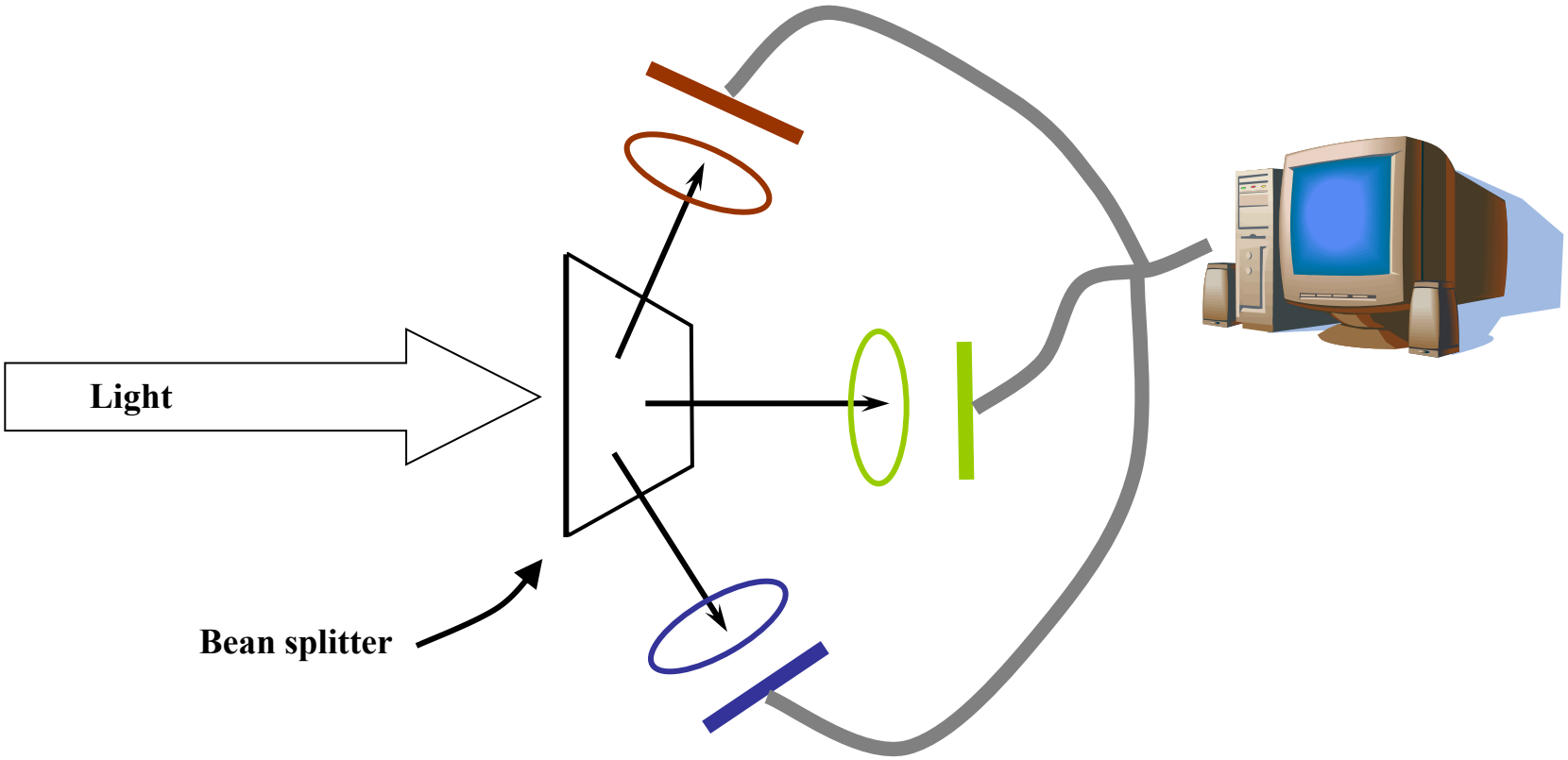
Property	CCD	CMOS
Image noise	comparable	comparable
electronics/pixel	Less	More (so what?)
Design complexity	Lower	Higher
Exposure	Global shutter	Global/Rolling
quantum efficiency	comparable	comparable
Potential frame rate	Lower	Higher
Power consumed	Much more	Much less
Fabrication cost	Much higher	Much lower
Maturity	comparable	comparable



In the last decade the gap in quality and maturity was closed, leaving CMOS imagers the preferred choice for inexpensive/low power/high speed imaging

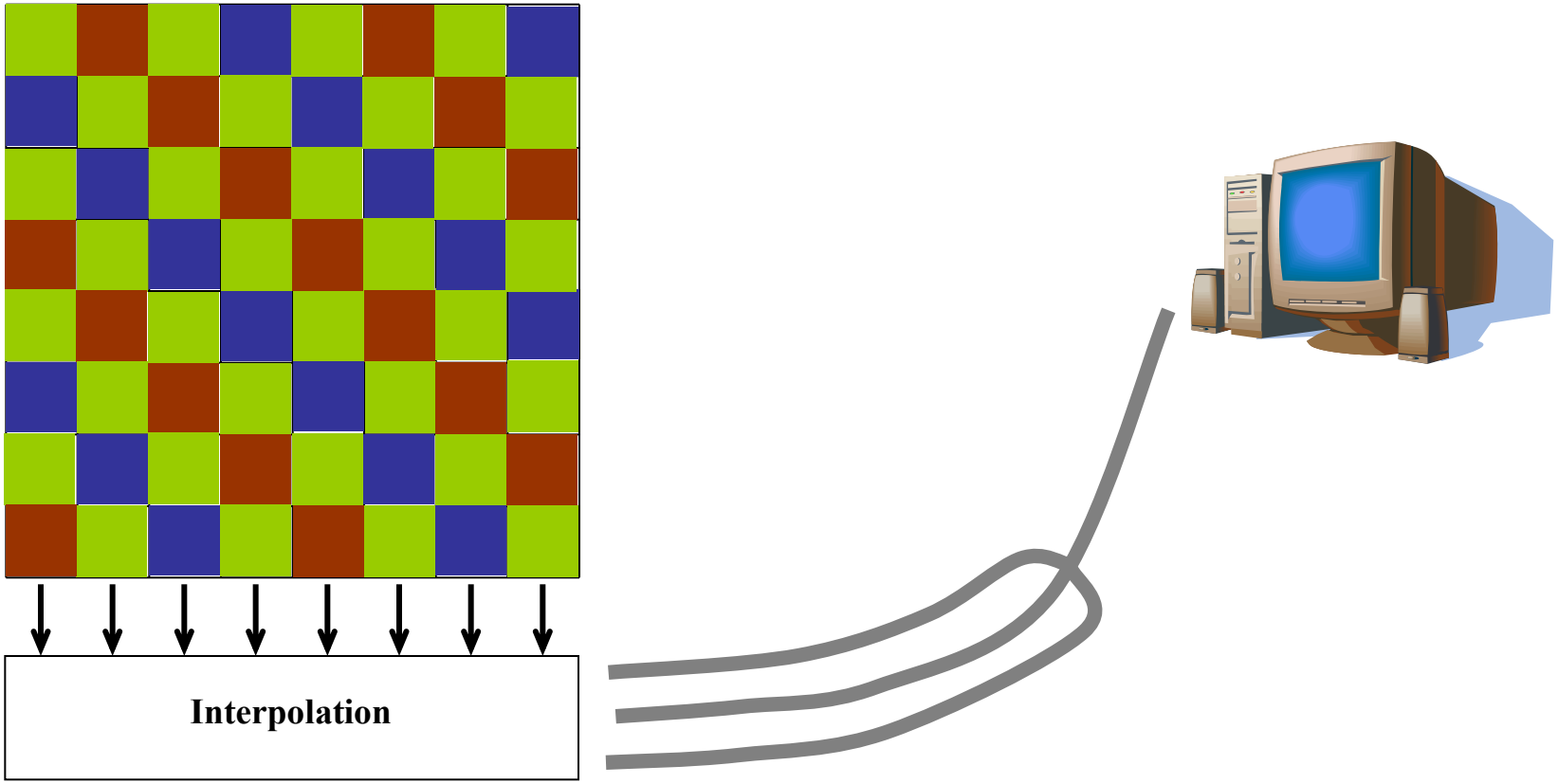
Digital light-capturing devices

Color 3-sensor devices

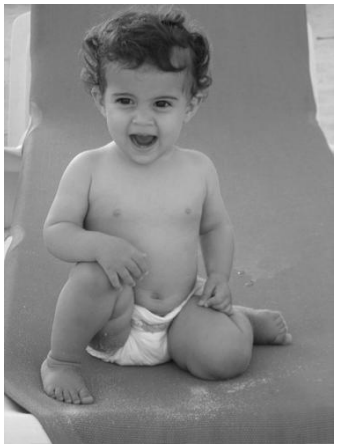


Digital light-capturing devices

Color single sensor devices – Bayer filters



Digital light-capturing devices



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124	124	110	93	96	78	38	42	66	62
197	196	199	200	196	200	199	195	203	201
150	144	125	119	118	108	144	164	227	81
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201	152	184	110	89	136	119	100	120	183
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203	179	174	173	138	117	100	107	107	118
188	145	111	169	160	135	107	74	59	63
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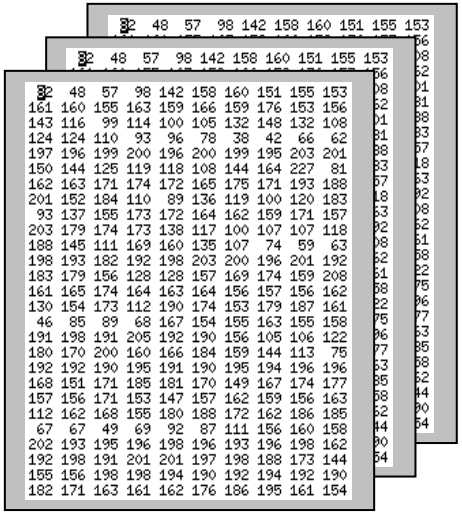
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